

Time limit: 20 minutes. Course staff will designate one problem in each section A–E. Complete all five designated problems. Show enough work to make your reasoning clear, define requested predicates and domains, and follow the directions for every lettered part. Approved testing accommodations apply.

A. Construct and interpret truth tables

Common directions for A.1–A.5. Construct a complete eight-row truth table. Show useful intermediate columns and classify the final proposition as a tautology, contradiction, or contingency.

A.1

$$p \rightarrow (\neg q \vee r)$$

A.2

$$\neg p \rightarrow (q \rightarrow r)$$

A.3

$$(p \rightarrow q) \vee (\neg p \rightarrow r)$$

A.4

$$(p \rightarrow q) \wedge (\neg p \rightarrow r)$$

A.5

$$(p \leftrightarrow q) \vee (\neg q \leftrightarrow r)$$

B. Translate meaning and diagnose translation errors

Common directions for B.1–B.5. Each problem has three parts: (a) translate conditional language into propositional logic; (b) translate between English and predicate logic; and (c) diagnose a proposed translation. Define every predicate and domain not already supplied. For part (c), say whether the proposal is correct, repair it if needed, and explain the error in one sentence.

B.1

- (a) Let e mean “you can edit a protected encyclopedia entry” and a mean “you are an administrator.” Translate: “You cannot edit a protected entry unless you are an administrator.”
- (b) Let $Q(x, y)$ mean “student x sent a message to student y .” Translate $\exists x \forall y Q(x, y)$ into ordinary English.
- (c) A student says the formula in (b) means “Every student sent a message to at least one student.” Diagnose the claim.

B.2

- (a) Let m mean “you can see the movie,” e mean “you are over 18,” and p mean “you have parental permission.” Translate: “You can see the movie only if you are over 18 or have parental permission.”
- (b) With $Q(x, y)$ as in B.1, translate $\forall x \exists y Q(x, y)$ into ordinary English.
- (c) A student translates “One student sent a message to every student” as $\forall y \exists x Q(x, y)$. Diagnose the translation.

B.3

- (a) Let g mean “you can graduate,” r mean “you completed your major requirements,” m mean “you owe the university money,” and b mean “you have an overdue library book.” Translate: “You can graduate only if you completed the requirements, owe no money, and have no overdue library book.”
- (b) Let $P(x, y)$ mean “student x has taken computer-science course y .” Translate $\exists y \forall x P(x, y)$ into English.
- (c) A student instead writes $\forall x \exists y P(x, y)$ for the sentence in (b). Diagnose the change in meaning.

B.4

- (a) Let w mean “you may use the airport wireless network,” d mean “you pay the daily fee,” and s mean “you subscribe to the service.” Translate: “To use the network, you must pay the daily fee unless you are a subscriber.”
- (b) Let $L(x, y)$ mean “person x loves person y .” Translate “Everybody loves somebody.”
- (c) A student claims that $\exists y \forall x L(x, y)$ is an equivalent translation of (b). Diagnose the claim.

B.5

- (a) Let p mean “the message is scanned for viruses” and q mean “the message came from an unknown system.” Translate: “It is necessary to scan a message whenever it comes from an unknown system.”
- (b) Let $S(x)$ mean “ x is a student,” $F(y)$ mean “ y is a faculty member,” and $A(x, y)$ mean “ x asked y a question.” Translate: “Some student has asked every faculty member a question.”
- (c) A student uses $A(y, x)$ in place of $A(x, y)$ in (b). Diagnose what the reversed argument order says.

C. Negate quantified statements

Common directions for C.1–C.5. For (a), define predicates, symbolize the statement, negate it with no negation immediately before a quantifier, and translate the result into ordinary English. For (b), rewrite the given formula so that every negation appears immediately before a predicate.

C.1

- (a) All dogs have fleas. The domain is all things.
- (b) $\neg \exists y \exists x P(x, y)$

C.2

- (a) No monkey can speak French. The domain is all things.
- (b) $\neg \forall x \exists y P(x, y)$

C.3

- (a) Some old dogs can learn new tricks.
The domain is all things.
- (b) $\neg\forall x\forall y P(x, y)$

C.4

- (a) No one in this class knows both French and Russian. The domain is all people.
- (b) $\neg\forall y\exists x P(x, y)$

C.5

- (a) All Swedish movies are serious. The domain is all things.
- (b) $\neg\forall y\forall x(P(x, y) \vee Q(x, y))$

D. Determine consistency and satisfiability

Common directions for D.1–D.5. Define one atomic proposition for each simple statement and translate every condition. Determine whether the conditions are consistent. Give a satisfying assignment when one exists; when none exists, give a short contradiction. Complete the additional balancing request in the final sentence of each problem.

D.1

The system is in multiuser state if and only if it is operating normally. If the system is operating normally, the kernel is functioning. The kernel is not functioning or the system is in interrupt mode. If the system is not in multiuser state, it is in interrupt mode. The system is not in interrupt mode. If the conditions are inconsistent, identify a minimal collection of the given conditions that already forces the contradiction.

D.2

Whenever system software is being upgraded, users cannot access the file system. If users can access the file system, they can save new files. If users cannot save new files, the software is not being upgraded. In addition to deciding consistency, list all satisfying assignments for the three atomic propositions.

D.3

The router can send packets to the edge system only if it supports the new address space. Supporting the new address space requires the latest software release. If the latest software release is installed, the router can send packets to the edge system. The router does not support the new address space. In addition to deciding consistency, determine which truth values are forced.

D.4

If the file system is not locked, new messages are queued. The file system is not locked if and only if the system is functioning normally. If messages are not queued, they are sent to the message buffer. If the file system is not locked, messages are sent to the buffer. Messages are not sent to the buffer. In addition to deciding consistency, explain which conclusions follow from the last condition.

D.5

Jasmine becomes unhappy if she attends a party with Samir. Samir attends only if Kanti attends. Kanti will not attend unless Jasmine attends. Determine which subsets of Jasmine, Samir, and Kanti can be invited without violating these conditions. State every satisfying combination, including the possibility of inviting no one.

E. Derive logical equivalences with cited rules

Permitted rules for E.1–E.5

Give a chain of equivalences from the left-hand expression to the right-hand expression. Cite exactly one rule at every step. Each rule may be used in either direction.

- **Implication:** $a \rightarrow b \equiv \neg a \vee b$.
- **Double Negation:** $\neg\neg a \equiv a$.
- **De Morgan:** $\neg(a \vee b) \equiv \neg a \wedge \neg b$ and $\neg(a \wedge b) \equiv \neg a \vee \neg b$.
- **Distributive:** $a \vee (b \wedge c) \equiv (a \vee b) \wedge (a \vee c)$ and $a \wedge (b \vee c) \equiv (a \wedge b) \vee (a \wedge c)$.
- **Associative:** $(a \vee b) \vee c \equiv a \vee (b \vee c)$ and $(a \wedge b) \wedge c \equiv a \wedge (b \wedge c)$.
- **Commutative:** $a \vee b \equiv b \vee a$ and $a \wedge b \equiv b \wedge a$.
- **Idempotent:** $a \vee a \equiv a$ and $a \wedge a \equiv a$.

Changing parentheses requires Associative; changing order requires Commutative.

E.1

Show that

$$(p \rightarrow q) \wedge (p \rightarrow r) \quad \text{and} \quad p \rightarrow (q \wedge r)$$

are logically equivalent.

E.2

Show that

$$(p \rightarrow r) \wedge (q \rightarrow r) \quad \text{and} \quad (p \vee q) \rightarrow r$$

are logically equivalent.

E.3

Show that

$$(p \rightarrow q) \vee (p \rightarrow r) \quad \text{and} \quad p \rightarrow (q \vee r)$$

are logically equivalent.

E.4

Show that

$$(p \rightarrow r) \vee (q \rightarrow r) \quad \text{and} \quad (p \wedge q) \rightarrow r$$

are logically equivalent.

E.5

Show that

$$\neg p \rightarrow (q \rightarrow r) \quad \text{and} \quad q \rightarrow (p \vee r)$$

are logically equivalent.